Rebalancing Gaussian Location Loss for High-Precision Detection on Remote Sensing Images

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***Abstract*— Aerial image objects are usually orientated arbitrarily, with a large scale range, and densely distributed. Traditional horizontal bounding box (HBB) detectors tend to filter out densely distributed objects leading to missed detections, such as ship (SH) and vehicle. Therefore, oriented object detection has become a mainstream solution in recent years. The 2-D Gaussian distribution representation of the oriented bounding boxes (OBBs) solves the problem of angular discontinuity and boundary discontinuity and thus gets more attention. However, as the aspect ratio of the object gradually decreases, its predicted angular performance continues to decrease. We find that the angular gradient of an object decreases sharply as the aspect ratio decreases, resulting in a large gradient gap between a small aspect ratio object (SARO) and a large aspect ratio object (LARO). It makes the detector prefer to ignore SARO during training, which weakens the high precision performance of SARO. We call this phenomenon shape imbalance. To solve the problem, we proposed a simple gradient rebalancing strategy named shape balance. Since the shape imbalance is only related to the aspect ratio of the object, we designed a modulation function with an inverse aspect ratio to calculate the balance coefficient. The principle of the function is that the larger the aspect ratio, the smaller the balance coefficient; the smaller the aspect ratio, the larger the balance coefficient. We aim to get the balance coefficients for objects with different aspect ratios. Location loss multiplied by a balance coefficient can directly adjust the gradient gap between objects with different aspect ratios to achieve a rebalancing effect. Extensive experiments conducted on DOTA-v1.0 dataset and DIOR-R dataset verify the effectiveness of our proposed method. Our method improves the detection performance of Gaussian location loss by an average of 2.08%/1.01%(AP_{75}/mAP) metrics on the DOTA-v1.0 dataset and 1.17%/0.82%(AP_{75}/mAP) improvements for DIOR-R dataset.**

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***Index Terms*— Aerial object detection, Gaussian distribution, oriented object detection, shape imbalance.**

I. Introduction

TARGETS in the remote sensing scene usually are distributed densely, span a wide range of scales, and have arbitrary orientations. Limited by the nonmaximum suppression (NMS) postprocessing method, the traditional horizontal bounding box (HBB) detector [13], [21], [22], [23], [25] is prone to suppress the valid detection of the surroundings in scenarios with densely distributed objects, causing missing detections and thus reducing the detector's performance. Oriented bounding box (OBB) detectors [1], [4], [6], [7], [8], [12], [14], [24], [27], [31], [33], [41] introduce angular information on the basis of HBB. OBB is closer to the actual boundary of the object than the HBB and has less background area in the bounding box. In an object distributed densely scenario, the OBBs of different objects have fewer overlapping regions, which makes the NMS produce significantly fewer missed detections, and thus, the OBB detectors show a far superior performance to the HBB detectors on many datasets [3], [16], [26]. From another perspective, precisely estimating the angle of objects will help people get high-quality information to benefit downstream tasks. Considering the many advantages of OBB over HBB, it is currently dominated by aerial image object detection. However, the periodically varying angle of the OBB generates the problem of angular discontinuity and boundary discontinuity, which had a strong negative impact on the earlier methods [6], [14], [31]. The angular discontinuity problem and the boundary discontinuity problem can disrupt the convergence process of the detector and prevent the model from learning the boundary information effectively. The long-edge (LE) definitions solve the boundary discontinuity problem but are still suffering from the angular discontinuity problem. The four-point representation and the vector representation avoid the above problem, but do not constrain the bounding box. CSL [30] encodes angles into categories, converting the prediction task into a classification task, but they are difficult to capture precise orientations and cannot measure angle differences. As the aspect ratio of the object continues to grow, the intersection over union (IoU) is more sensitive to angular deviations. To address the above problem, Yang et al. [32] designed a novel localization loss paradigm,

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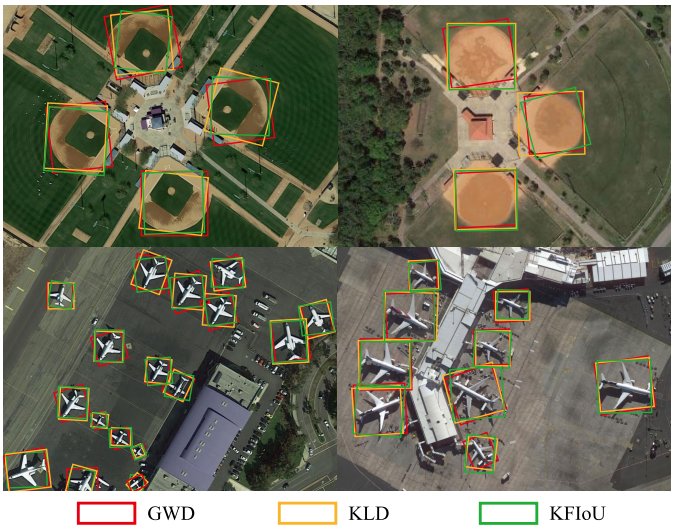


Fig. 1. Detection results of Gaussian distribution loss on SAROs (such as airplanes and BDs). The detector is R^3Det, where different colored bounding boxes show that different losses were used. For these two types of square-like objects, the angle of the detection box tends to deviate significantly, making it challenging to get the correct orientation.

where they converted the OBB into a 2-D Gaussian distribution, and estimated the distance between OBBs by calculating the distance between probability distributions, and thus, they called Gaussian distance loss (GDL). GWD [32] uses the Wasserstein distance; KLD [34] uses the Kullback–Leibler divergence; ProbIoU [17] uses the Hellinger distance; and KFIoU [36] applies Kalman filtering (or the Gaussian multiplication theorem) to estimate IoU. GDL constructs the distance metric from a global perspective by coupling the parameters of the bounding box, which achieves better results than the multiparameter independent loss scheme that focuses on the local perspective. The GDL's parameter-coupled form makes large aspect ratio objects (LAROs) dominate the gradient, leading the detector to precisely estimate the boundaries of the LAROs [34]. It also results in a significant offset in the angular estimation of the small aspect ratio objects (SAROs). SAROs are much less sensitive to angular deviation, which does not affect the evaluation results with the mainstream AP_{50} (AP indicates average precision and 50 indicates that the IoU threshold is 0.5) metrics, and this is the main reason the gap between GDL and L1/Smooth L1 loss is smaller with the AP_{50} metrics. The work of Li et al. [11] shows a significant detection performance decrease in SAROs for GDL versus L1/Smooth L1 loss at high-precision metrics such as AP_{75}. As shown in Fig. 1, we applied three GDLs (GWD [32], KLD [34], and KFIoU [36]) to the R^3Det [31] detector and visualized for airplanes and baseball diamonds (BDs). It is difficult to predict the correct angle of airplanes and BDs using these GDL detectors, which is in strong contrast to their excellent performance on objects with large aspect ratios, such as vehicles and ships (SHs). This means that the GDL presents two different detection performances on SAROs and LAROs. For GDL, as the object aspect ratio gradually approaches 1, the angular gradient at the same angular difference rapidly tends to 0. It prevents SAROs from learning efficiently due to too small a gradient. The phenomenon is affected by object aspect ratio, so we call it shape imbalance. The shape imbalance causes the detector to be insensitive to the rotation information of SAROs, which has little effect on AP_{50} but has a greater impact on high-precision metrics. However, high-precision OBB detection plays a critical role in improving the quality of remote sensing image interpretation. Therefore, shape imbalance is an urgent problem to be solved.

Motivations: Both the GDL loss and gradient of the angle for SAROs are much smaller than for LARO, thus affecting the model's preference and making it more likely to ignore the angular information of SAROs. To explore this shape-related phenomenon, we calculated the gradient ratios between objects with different aspect ratios. Fig. 2 illustrates the effect of the three GDLs on the gradients of objects with different aspect ratios. In classical losses, such as GWD [32], KLD [34], and KFIoU [36], the angular gradient of an object with 4.0 aspect ratio is over 1000 times higher than a 1.05 aspect ratio object.

GDL greatly enhances the gradient of LAROs, significantly improving its precision, but SAROs show a detection performance decrease in high-precision metrics such as AP_{75}. The learning efficiency of gradient backpropagation-based models relies on the update step length. A suitable and efficient update step length for LAROs is difficult to learn the boundary information of SAROs after weakening thousands of times. We believe that the huge gradient gap between objects with different aspect ratios is the main reason for the phenomenon of shape imbalance. It weakens GDL, and thus, we design a method to rebalance the gradients of objects, makes the detector pay more attention to SAROs, which are used to improve the localization precision of SAROs.

Our contributions are as follows.

1) We analyze the reason that GDL causes large angular deviations for the SAROs under general conditions and summarize it as a shape imbalance problem.
2) We propose a general loss rebalancing strategy for increasing the loss of SAROs and making the model more attentive to the angles of SAROs.
3) We have conducted extensive experiments to verify the validity of the shape balance theory. Modified GDL improves the high-precision performance of the detector.

II. Related Works

A. Nonprobability Distribution Localization Loss

OBB has many advantages over HBB, but the localization task becomes more complex as angular information is introduced. The OBB localization loss is usually designed as an extended form of the HBB localization loss, such as L1/Smooth L1 loss, IoU loss, and GDL. Many problems exist with the direct regression of the L1/Smooth L1 loss on the OBB parameters.

1) Angle periodicity yields angle discontinuities, boundary discontinuities, and the dichotomous representation of square objects.
2) Angle info is not independent of position and scale. Objects with larger aspect ratios are more sensitive to angles.

For example, the boundary of an OpenCV (OC)-defined object swaps around a critical angle, and an LE-defined object cannot uniquely determine a square object. The same angular

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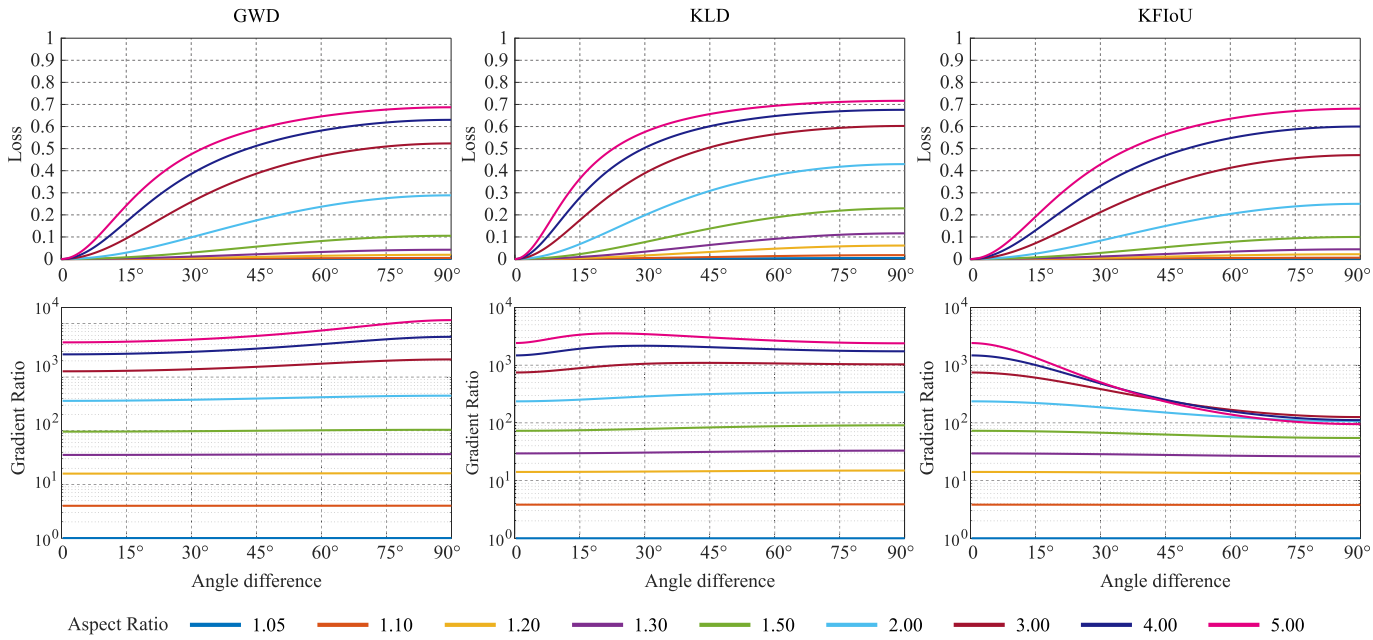


Fig. 2. Comparison of different aspect ratio objects on GDL and their gradients. Both loss and gradient are calculated by adjusting the angular difference between two objects with the same scale and center point. We further calculate the ratio between the gradients to visualize the gap between them. All gradients are divided by the gradient when the aspect ratio is 1.05, so it has a constant gradient ratio of 1. Classical GDL has large gaps in gradients for objects with different aspect ratios.

deviation makes the IoU more variable for larger aspect ratio objects. The IoU loss is differentiable for HBB, but it is difficult for OBB. To solve this problem, PIoU simplifies the process by converting the calculation of the overlap area between objects into a grid point count. APE [44] and H2RBox v2 [40] simulate IoUs by projection between OBBs. While P2P [37] uses multiple triangles to calculate the distance between two OBBs. CSL [30] and DCL [29] discretize the angles and replace the continuous value prediction task with classification, but their accuracy is limited by the degree of discretization. PSC [39] uses phase-shift algorithms to encode angles, which solve the angle discontinuity problem. However, its prediction results are directly decoded into angles, which lack theoretical support. Some methods no longer restrict the shape of the OBB, and they use multiple vectors [15], [38], [42], multiple vertices [5], [28], [45], or point set [9], [10], [18] to define the OBB.

B. Probability Distribution Localization Loss

Unlike the previous OBB location loss, GDL does not directly evaluate the difference between the two OBBs. It converts the OBB into a 2-D probability distribution and represents the distance between OBBs in terms of the distance between the two distributions. The probability distribution and the OBB are spatially aligned, so the distance between the distributions can be thought of as a measure of OBB distance. GWD [32], KLD [34], [34], ProbIoU [17], and KFIoU [36] are all models by Gaussian distributions, and the difference between them is the distribution distance method used. GWD adopts the Wasserstein distance, which couples the scale and angle of an object for regression, but the loss is not scale-invariant. KLD uses Kullback–Leibler divergence to compute the loss. Compared to GWD, KLD satisfies scale invariance but does not satisfy the exchange law. ProbIoU is a GDL using the Hellinger distance, which is practically irrelevant to IoU. KFIoU differs from the above three methods. It no longer works with probability distribution distances to calculate losses. KFIoU uses Kalman filtering to get the Gaussian distribution parameters of the intersecting region and compute its area further to get the loss. The KFIoU metric consists of the center point loss and IoU loss. It is worth noting that the center point loss lacks normalization, thereby resulting in a lack of scale invariance for KFIoU. The GDL based on Gaussian distribution has parametric coupling, which automatically adjusts the loss and gradient according to the predictions, and is better than L1/Smooth L1. However, the 2-D Gaussian distribution has a systematic deficit that leads to the angle missing problem. Li et al. [11] analyzed the theoretical reasons for the formation of the angular missing problem and introduced the Lamé curve to change the Gaussian distribution to resolve the angular missing problem. SGKLD [11] is an extension of KLD that represents a cluster of localization losses. When the parameter n of SGKLD is equal to 2, it is KLD.

III. Rethinking GDL

Compared with the traditional OBB localization loss, GDL has many advantages and has achieved impressive results on several remote sensing datasets. The classical GDL [17], [32], [34], [34], [36] is all predicated upon Gaussian distributions. The 2-D Gaussian distribution that corresponds to a nonsquare OBB is elliptical, whereas the distribution specific to a square object is circular. The ellipse is anisotropic and has a specific orientation so the model can learn the object's angle. However, the circle is isotropic and incapable of decoding actual angle information. After converting a square OBB into a 2-D Gaussian distribution, its orientation information is lost, resulting in the angle missing problem. To overcome this issue,

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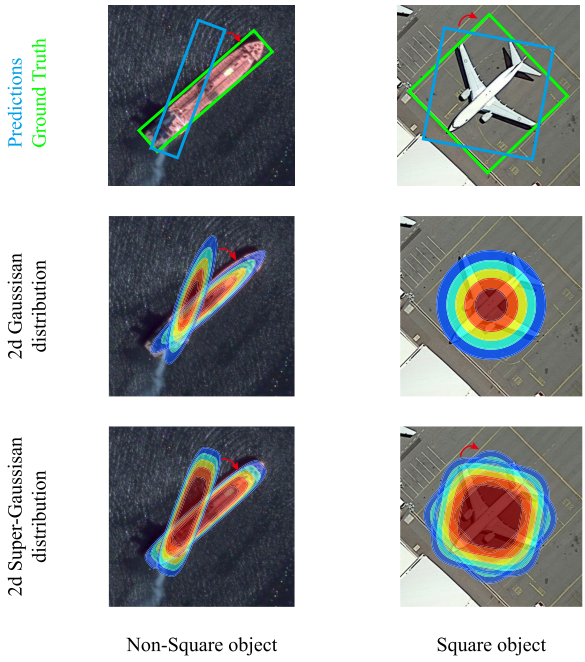


Fig. 3. GDL and SGKLD in OBB regression tasks. GDL converts the OBBs to Gaussian distributions and computes the probability distance between the distributions to model the distance between two OBBs. However, its effect on objects with different aspect ratios varies greatly. Both GDL and SGKLD can accurately estimate distances between objects with large aspect ratios. However, GDL is not able to learn the angle information of square objects effectively.

SGKLD [11] developed a super-Gaussian distribution utilizing Lamé curves, which keeps anisotropy in all conditions, effectively addressing the angle missing problem. Fig. 3 illustrates the angle missing problem faced by Gaussian OBB and the core idea of SGKLD. Although the issue of missing angles is a native bug of the Gaussian representations, the triggering conditions are strict and have a limited impact on general aerial object detection.

However, as we study the high-precision metrics of GDL, the results of GDL on SAROs (such as plane (PL) and BD) are weaker than other types of location loss. With the AP_{50} metric, many SAROs with large angular deviations are recognized as correct detections, while LAROs maintaining the same angular difference are easily identified as invalid detections. Angle variations can actually bring about drastic changes in IoU for LARO. However, the IoU reduction due to rotation is not severe for SARO and is well below the requirements of mainstream metric AP_{50}. While the AP_{50} metrics for GDL are excellent, the high-precision metrics on objects with small aspect ratios such as airplanes and BDs are severely degraded.

GDL has a high level of parameter coupling. They are usually optimized as a whole and dynamically adjust the losses and gradients according to the current predictions. The Gaussian distribution of SARO is still anisotropic. Thus, the GDL also optimizes the angle of SARO. What is the reason for the gap between SARO and LARO? Since GDL uses the same process for both SARO and LARO, the only possible reasons for the gap between them are imbalances in losses or gradients. We conducted a simple comparison of the angular gradients of GWD, KLD, and KFIoU, as shown in Fig. 2. The angular gradient of the scene was fixed as two OBBs with the same center and scale but at different angles. We selected objects with aspect ratios of {1.05, 1.1, 1.2, 1.3, 1.5, 2.0, 3.0, 4.0, 5.0} and calculated both their losses and angular gradients. To observe the differences in gradients for objects with varying aspect ratios, we divided all the angular gradients by the gradient at 1.05 to get the gradient ratio between them, i.e., the four subplots in the second row of Fig. 2. GWD, KLD, and KFIoU generate angular gradients on objects with an aspect ratio of 4 that is 800–2000 times greater than at 1.05. This huge gradient gap makes the update step for SARO's angle almost negligible. It means that the model tends to ignore the angle information of the SARO when it is updated. Because the aspect ratio of the object produces this gap, we call the phenomenon shape imbalance. For some non-GDL, the shape imbalance is absent. For example, when using the L1/Smooth L1 loss for individual angle optimization, the shape of the objects does not affect the angular gradient, so the gradient ratio is 1. Li et al. [11] compared the high-precision performance of L1/Smooth L1 and GDL on SAROs. The results showed that L1/Smooth L1 loss was better, while GDL only regressed to the center point of the SAROs, but could not learn the actual angle. In summary, we believe that the imbalance in shape is the primary cause of the weak detection performance of GDL on SARO. Rebalancing the ratios is necessary to enhance their detection performance.

IV. Proposed Methods

A. Shape Balance Function

Considering the generality of the shape balance function, it is not a good idea to design the balance function inside the loss function. The aim is not to design a new loss function, but to provide an effective remodeling solution. This balancing scheme increases the gradient of the SARO, so we propose to apply the shape balance function directly to the loss function. In this article, we multiply the balance coefficient and the localization loss directly to get the new localization loss

$$\text{Loss}_{\text{loc|new}} = f\text{Loss}_{\text{loc|old}} \tag{1}$$

where Loss_{loc} denotes the localization loss and f denotes the balance function.

We get the balance coefficient through the function, which is just a constant for the localization loss. Thus, the multiplication operation narrows the gap between objects with different aspect ratios. The shape imbalance is generated by the aspect ratio of the object, so the balance function should also be controlled by the aspect ratio k. The basic principles are as follows.

1) The larger the k is, the smaller the f is.
2) The smaller the k is, the larger the f is.
3) The lower bound of the function is 1.

Therefore, we designed the original balance function

$$f = 1 + \frac{1}{k-1}, k \geqslant 1 \tag{2}$$

where k indicates the aspect ratio of the object, and it is obtained by dividing the LE by the short edge of the object. Therefore, k must be greater than or equal to 1.

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To limit the influence of the balance coefficients, we add scaling functions to assist f. It determines the scaling factor according to the IoU between the prediction and ground truth. The range of the scaling factor can be defined as [0, 1]. The scaling function s is defined as

$$s = \mathrm{IoU}^n \tag{3}$$

where n is the hyperparameter, which controls the shape of the curve. The full balance function is

$$f^* = 1 + \frac{\mathrm{IoU}^n}{k-1}, \quad k \geqslant 1. \tag{4}$$

We multiplied the balance coefficient computed by (4) with the localization loss of the object and got the new localization loss.

B. Function Analysis

Equation (2) balances the gradients of objects with different aspect ratios, but it introduces two problems.

1) The balance coefficient of a square object is infinity, resulting in a model that cannot be iteratively updated.
2) Model learning is a process from disorder to order; is it appropriate to apply the coefficients determined by the aspect ratio to the whole process of training?

For issue 1, we think it is sufficient to set an upper bound, which does not affect the angles. Because the GDLs suffer from the angle missing problem, and they cannot solve the problem themselves. However, the balance coefficient still affects the gradient of the center point and the scale. For issue 2, we believe that the balance function should not work for the whole process during training.

The shape imbalance phenomenon occurs when the centroid and scale are close to the target. Applying the balance function throughout means that the gradients at the centroids and scales of SARO may be strengthened in the early stages of training, which disturbs the model convergence. The IoU between model prediction and ground truth is a popular metric in target detection tasks. It evaluates the distance between two bounding boxes. When IoU is low, there is a large gap between model predictions and ground truth, and the model primarily focuses on the center coordinates and scale of the SARO. When IoU is high, we believe that the shape imbalance plays a major role, and it prevents the SARO from learning the correct angle further. It results in the GDL having a much lower high-precision metric (such as AP_{75}) on SARO than the L1/Smooth L1 loss. Via an IoU-related function that determines the scaling factor of the current sample, we can restrict the scenarios in which shape balance can be used. The function should be monotonically increasing and the range should be as close to [0, 1] as possible. Therefore, we designed the scaling function in (3) and got the final balance function f^* in (4).

V. Experiments

A. Datasets

DOTA [26] is a large-scale dataset for aerial object detection. The data are collected from different sensors and platforms. DOTA-v1.0 contains 2806 large aerial images with size ranges from 800 × 800 to 4000 × 4000 and 188 282 instances among 15 common categories: PL, BD, bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), SH, tennis court (TC), basketball court (BC), storage tank (ST), soccer-ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC). We used both the train and validation sets for training and used the test set for testing. All images were cropped into 1024 × 1024 patches with a gap of 200 for single scale and 500 (gap) for multiscale. The multiscale arguments were {0.5, 1.0, 1.5}. We also applied random flipping and the random rotation argument method during training.

DIOR-R [3] is a large-scale benchmark dataset for object detection in optical remote sensing images, which consists of 23 463 images and 192 518 object instances annotated with OBBs. It consists of small, cropped images of 800 × 800 pixels, unlike the DOTA dataset. DIOR-R dataset covering a wide range of scenes and 20 common object classes. The categories of objects in DIOR-R include airplane (PL), airport (APO), baseball field (BF), BC, BR, chimney (CH), expressway service area (ESA), expressway toll station (ETS), dam (DAM), golf field (GF), GTF, HA, overpass (OP), SH, stadium (STA), ST, TC, train station (TS), vehicle (VE), and windmill (WM).

B. Implementation Details

We use MMRotate [43] for implementation. We use single Nvidia RTX-3090 GPUs at 24G with a total batch size of 2 for training and testing. All datasets are trained by 12 epochs, and the learning rate (LR) is reduced tenfold at eight epochs and 11 epochs, respectively. The initial LR is 0.0025 and adopts stochastic gradient descent (SGD) to optimize the model. The momentum and weight decay were 0.9 and 0.0001, respectively. The NMS threshold was set to 0.1 when merging image patches, and the confidence threshold was set to 0.05 during testing. During training, we do not change the base weights or coefficient for location loss in the baseline.

C. Ablation Experiments

We performed ablation experiments on the DOTA-v1.0 dataset for two architectures of detectors (RetinaNet-O [13] and R^3Det [31]) and three GDLs (GWD [32], KLD [34], [34], and KFIoU [36]). RetinaNet-O [13] and R^3Det [31] adopt ResNet50 as the backbone. We use the training set for training and the validation set for testing. All results are shown in Tables I–III, where baseline indicates that the loss does not use the shape balance strategy. n is the hyperparameter in (4). When $n = 0$, (4) degenerates into (2). The bold value means the best result.

From Tables I–III, it can be observed that introducing the shape balance strategy early in the model's training greatly reduces the detection performance of the detector [see (2)]. This is the same as our speculation, and the detector performance will continue to improve and exceed the baseline as n increases. Predictably, as n continues to increase, f^* will be almost equal to 1, indicating that the detection performance of the detector should be close to the baseline. It can be described as $n \to \infty \Rightarrow f^* \to 1 \Rightarrow \mathrm{Det}_{\mathrm{balance}} \to \mathrm{Det}_{\mathrm{baseline}}$.

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TABLE I
IMPACT OF PARAMETER n ON GWD LOSS

Detectors	Shape Balance	Plane			Baseball Diamond			Total		
		AP_{50}	AP_{75}	mAP	AP_{50}	AP_{75}	mAP	AP_{50}	AP_{75}	mAP
RetinaNet-O[13]	baseline	76.32	43.05	43.89	65.98	17.78	29.75	54.46	27.85	29.55
	n=0	76.31	33.72	40.33	57.56	18.39	29.21	48.07	23.33	25.64
	n=2	76.18	38.57	43.13	58.71	19.14	30.20	52.46	26.43	28.90
	n=4	76.21	41.76	44.51	65.58	19.80	30.79	53.58	27.73	29.55
	n=6	76.78	43.06	**46.11**	67.27	18.45	31.11	**55.21**	28.65	30.43
	n=8	**77.37**	42.96	45.02	**69.38**	20.86	33.42	55.06	28.52	30.47
	n=10	76.70	**44.32**	45.82	67.10	21.38	**33.87**	54.54	**28.73**	**30.93**
	n=12	75.88	43.07	44.50	67.02	**21.52**	32.61	54.96	28.13	30.40
R^3Det[31]	baseline	77.39	42.37	45.09	**71.48**	15.68	32.18	57.85	28.87	31.56
	n=0	77.61	31.86	42.40	62.02	22.40	32.31	54.02	24.27	28.52
	n=2	78.09	39.77	43.90	66.15	21.34	32.32	55.83	27.71	30.19
	n=4	77.04	40.35	44.37	68.67	21.54	34.12	58.22	28.96	31.75
	n=6	77.61	41.98	45.16	68.17	22.40	32.88	58.06	29.52	31.84
	n=8	**78.23**	42.86	**46.84**	71.30	**23.83**	**35.19**	57.20	30.24	32.14
	n=10	77.62	**42.98**	46.54	67.68	22.66	33.06	**58.55**	30.19	**32.58**
	n=12	77.51	42.72	46.70	67.43	22.87	33.95	57.94	**30.43**	32.51

TABLE II
IMPACT OF PARAMETER n ON KLD LOSS

Detectors	Shape Balance	Plane			Baseball Diamond			Total		
		AP_{50}	AP_{75}	mAP	AP_{50}	AP_{75}	mAP	AP_{50}	AP_{75}	mAP
RetinaNet-O[13]	baseline	73.89	40.59	41.37	66.75	15.98	30.29	54.74	27.52	29.65
	n=0	74.43	32.78	39.45	49.97	16.16	22.61	44.84	21.96	24.22
	n=2	75.77	41.51	43.21	64.74	20.19	32.50	51.70	26.68	28.41
	n=4	76.26	43.21	44.97	59.43	20.20	30.63	52.69	28.53	29.70
	n=6	76.31	44.49	45.20	65.93	22.48	31.52	53.94	**29.40**	30.32
	n=8	76.42	**49.24**	**46.03**	66.86	**23.55**	33.69	54.77	29.14	**30.63**
	n=10	76.22	43.36	45.30	68.18	21.92	**34.29**	**55.32**	28.61	30.48
	n=12	**76.74**	43.96	45.68	**68.63**	20.63	33.27	55.16	28.33	30.19
R^3Det[31]	baseline	75.95	42.72	44.02	**68.07**	20.09	31.86	56.74	28.87	31.35
	n=0	72.80	31.48	37.62	47.15	12.80	21.64	46.62	21.61	23.92
	n=2	**77.70**	41.11	45.63	62.75	20.90	31.26	54.97	27.79	30.31
	n=4	77.12	42.91	**46.53**	61.89	21.56	30.96	55.94	29.74	31.41
	n=6	76.59	41.93	45.57	65.47	21.60	33.22	56.37	29.71	31.28
	n=8	77.55	**43.29**	46.37	65.26	22.76	33.14	56.69	30.18	31.63
	n=10	77.50	43.01	46.32	64.59	23.57	**34.10**	56.83	**30.67**	**32.29**
	n=12	76.86	42.00	45.44	63.46	**24.69**	32.62	**57.39**	29.66	31.94

TABLE III
IMPACT OF PARAMETER n ON KFIOU LOSS

Detectors	Shape Balance	Plane			Baseball Diamond			Total		
		AP_{50}	AP_{75}	mAP	AP_{50}	AP_{75}	mAP	AP_{50}	AP_{75}	mAP
RetinaNet-O[13]	baseline	77.85	32.88	41.19	**69.17**	11.56	28.95	**56.66**	22.01	27.70
	n=0	76.81	32.73	40.13	63.11	15.26	28.28	51.87	22.57	26.19
	n=2	**78.54**	**34.99**	**42.11**	64.32	**23.78**	32.65	55.05	24.09	27.94
	n=4	77.60	34.45	42.00	67.65	21.72	32.86	55.86	**25.38**	**29.08**
	n=6	77.20	34.08	41.90	65.43	16.30	31.73	54.55	24.74	28.63
	n=8	75.67	33.06	41.40	67.31	21.33	**33.48**	54.78	24.98	28.59
	n=10	78.23	34.22	41.88	64.88	13.65	29.11	55.55	23.77	28.22
	n=12	75.04	32.49	39.64	65.29	16.93	32.50	53.99	24.20	28.18
R^3Det[31]	baseline	75.07	32.41	39.85	63.98	15.92	29.53	54.72	25.67	28.82
	n=0	**78.76**	31.69	42.09	67.16	13.96	30.85	56.01	23.20	28.24
	n=2	78.07	32.94	**42.95**	65.39	17.42	31.14	57.43	25.20	29.94
	n=4	77.02	33.31	41.94	67.02	**19.98**	**33.07**	56.99	**26.90**	**30.51**
	n=6	77.86	**33.69**	42.03	65.67	19.71	31.15	57.20	26.11	30.09
	n=8	77.18	33.41	42.45	63.96	17.76	29.58	57.27	26.18	30.01
	n=10	77.33	33.42	41.81	**71.10**	16.53	32.07	**57.85**	25.22	29.65
	n=12	76.75	32.88	41.21	63.89	16.92	29.96	57.45	25.08	29.31

1) Effect of n: For more visualization, we subtracted the baseline from the precision in Tables I–III and obtained Figs. 4 and 5, where *green block* represents the detection performance improvement, while *orange block* represents the detection performance degradation. Figs. 4 and 5 illustrate the shape balance results for RetinaNet-O and R^3Det with multiple GDLs, respectively.

The optimal n is different for various detectors and losses, but they all maintain similar trends. When $n = 0$, there is a significant decrease in the accuracy of the model, and

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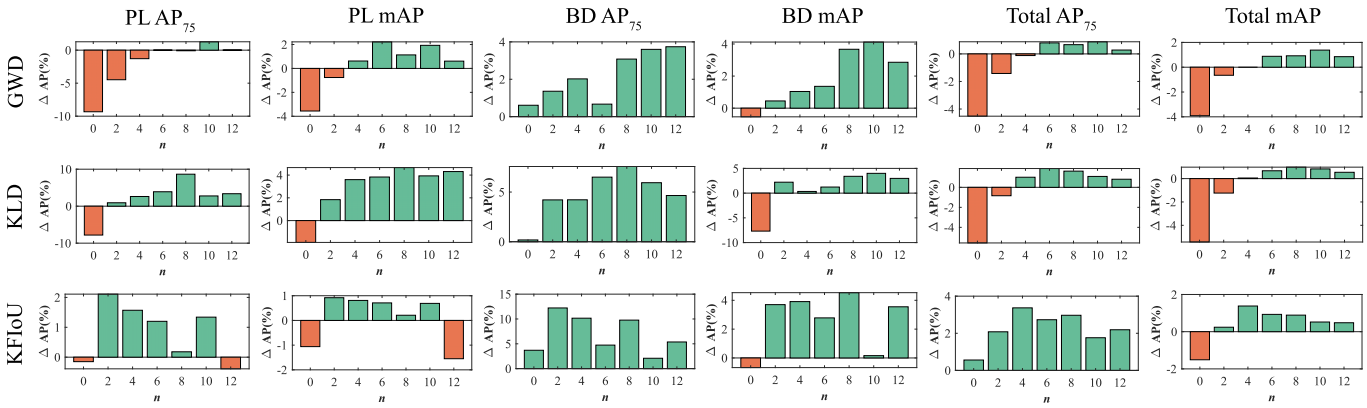


Fig. 4. Effect of parameter n on the detection performance of the RetinaNet-O detector, where the horizontal axis denotes the value of parameter n in the range [0, 12] and the vertical axis denotes the value of the current metric reduced by the baseline. The red block and green block denote the detection performance decrease and increase, respectively. PL and BD denote the plane and baseball diamond, respectively.

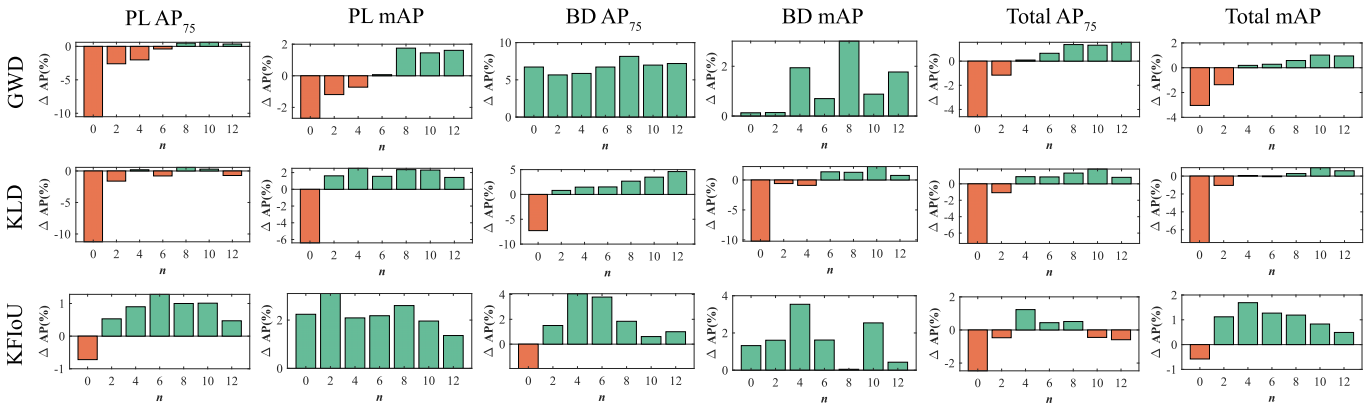


Fig. 5. Effect of parameter n on the detection performance of the R^{3}Det detector. Symbol meanings and legends are the same as in Fig. 4.

as n continues to increase, it will outperform the baseline and gradually converge to the baseline after achieving the peak. When the parameter n is small, the overall detection performance of the detector almost always decreases at some level, such as $n = 0$ and $n = 2$.

The phenomenon of shape imbalance is mainly affected by the aspect ratio of the target, but rebalancing the gradients between objects with different aspect ratios early in training tends to decrease the model's overall detection performance. The parameter n is essentially the influence of IoU on the balance coefficient. When $n = 0$, IoU does not affect the balance coefficients, and a larger n indicates a larger impact of IoU on the balance coefficients. In general, the IoU is highly correlated with the training procedures of the model. At the early stage of training, the IoU between prediction and ground truth is low, but with continuous iterations and parameter updates, the IoU between them gradually rises. Therefore, IoU can reflect the degree of convergence of the model and the training phase. The parameter n then serves as a BR between the balance coefficient and the training phase. Smaller n results in a detector performance decrease, implying that rebalancing should be applied later in training. But the larger the n is, the smaller the impact of rebalancing is. Through experiments on two detectors and three GDLs shown in Figs. 4 and 5, we believe that when n is within [6, 10], it allows for a steady improvement on model performance.

2) Small Aspect Ratio Object: The AP_{75} metrics for PL objects were not stable, but its mean average precision (mAP) metrics were significantly increased. Overall, shape balance produces a more stable improvement on the BD. The different n gained a boost of basically around 2%. By analyzing the gradient differences of objects with different aspect ratios, we discovered the widespread shape imbalance phenomenon of GDL and explained the reason why GDL could not effectively learn the angle information of SARO. The extensive experiments in Figs. 4 and 5 confirm that applying shape balance to GDL can effectively improve the high-precision metrics of SARO while improving the overall model performance.

3) GDL: From Figs. 4 and 5, there is a difference in the effect of shape balance on the loss of different GDLs.

For GWD, the optimal range of n is [8, 12], where both the overall and SARO high-precision metrics are improved. It has an overall detection performance improvement of about 0.95%, and the average detection performance improvement of SARO is 2.12%.

For KLD, the range of n is the same as for GWD, which has an overall detection performance improvement of about 0.69% and a SARO improvement of about 2.80%.

For KFIoU, the optimal range of n is [4, 10], and its overall detection performance improvement is about 1.09%, while SARO improves about 1.90%.

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TABLE IV
COMPARISON OF SHAPE BALANCE ON DOTA-V1.0 TRAINVAL/TEST SET (∗ MEANS USING SHAPE BALANCE, AND MS MEANS MULTISCALE TRAINING AND TESTING)

Detector	Method	MS	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HE	AP_{50}	AP_{75}	mAP
RetinaNet-O[13]	KLD	×	89.26	79.97	41.70	65.01	78.77	63.77	77.62	90.79	81.99	81.67	56.24	63.79	58.58	66.66	40.63	69.10	37.30	38.83
	KLD∗	×	89.04	77.89	41.58	67.22	78.15	63.28	77.67	90.89	80.15	82.78	58.28	63.88	57.68	66.92	44.75	69.34	38.50	39.55
	GWD	×	88.52	79.59	41.63	68.78	77.44	61.45	77.46	90.85	82.49	81.54	58.15	65.79	58.03	65.69	36.21	68.91	35.45	37.95
	GWD∗	×	88.25	79.80	42.11	67.94	78.08	62.63	77.58	90.82	81.62	80.62	55.73	64.33	58.07	65.38	45.47	69.23	36.97	38.73
	KFIoU	×	77.35	75.46	40.30	61.81	59.44	45.64	62.92	83.17	72.50	79.33	54.08	65.02	53.09	61.77	50.63	62.83	20.88	29.10
	KFIoU∗	×	82.72	75.68	42.39	59.13	61.38	46.50	64.53	83.89	70.92	79.71	54.92	63.42	53.07	59.33	47.16	62.98	25.24	30.94
	KLD	✓	89.67	83.61	48.51	76.05	76.92	73.84	84.98	90.63	84.71	86.95	68.50	71.84	64.32	71.87	64.95	75.82	46.59	45.56
	KLD∗	✓	89.14	84.97	48.57	78.44	77.48	73.24	85.19	90.36	83.30	85.89	70.96	69.18	64.53	73.01	68.07	76.16	48.63	46.51
	GWD	✓	89.24	83.63	48.65	75.77	76.60	72.43	84.53	90.27	85.38	85.23	69.23	69.42	64.33	72.15	67.29	75.61	47.11	45.48
	GWD∗	✓	89.17	80.67	47.86	76.17	76.78	72.96	84.58	90.46	84.27	84.82	69.36	68.32	64.38	72.03	64.05	75.06	48.47	46.20
	KFIoU	✓	89.83	83.47	49.40	77.75	76.01	72.04	83.16	90.46	85.69	85.09	70.12	71.29	63.12	72.95	66.21	75.77	38.94	41.55
	KFIoU∗	✓	89.62	83.49	49.48	71.16	75.50	72.03	84.68	90.24	84.89	86.10	67.70	70.42	63.93	72.87	72.41	75.63	41.61	42.79
R^3Det[31]	KLD	×	89.44	77.51	47.84	71.97	78.53	74.70	86.76	90.90	83.78	83.73	59.83	61.65	65.22	66.99	46.59	72.36	38.79	40.42
	KLD∗	×	89.50	76.94	48.44	69.14	78.55	75.16	87.00	90.88	85.27	83.94	64.50	61.50	65.18	67.46	46.87	72.69	40.47	41.71
	GWD	×	89.03	76.04	45.83	73.63	77.99	75.07	85.39	90.90	82.45	84.53	60.16	61.11	65.63	68.24	51.98	72.53	37.88	39.90
	GWD∗	×	89.48	76.30	47.08	72.85	77.56	74.02	85.82	90.90	83.01	83.97	61.94	63.15	64.62	68.33	51.12	72.68	38.58	40.66
	KFIoU	×	89.56	79.44	45.38	70.38	77.70	73.34	83.57	90.88	83.13	84.26	58.19	64.07	63.11	65.15	45.05	71.55	31.75	37.06
	KFIoU∗	×	89.55	77.99	44.76	67.72	78.01	74.49	84.42	90.90	81.33	83.97	60.58	64.91	63.41	66.39	52.62	72.07	34.33	38.31
	KLD	✓	89.33	83.09	55.20	83.44	80.22	84.09	88.64	90.84	84.84	87.62	70.36	71.66	76.23	80.84	64.76	79.41	50.94	48.44
	KLD∗	✓	89.35	83.35	55.85	82.30	80.33	84.54	88.59	90.86	84.97	87.88	74.27	70.04	75.75	79.22	68.28	79.71	52.45	49.43
	GWD	✓	89.24	84.04	55.16	82.96	80.04	84.22	88.82	90.85	86.60	87.86	69.99	71.86	75.59	78.46	67.47	79.54	49.11	48.39
	GWD∗	✓	89.05	82.39	55.31	81.23	80.04	84.75	88.59	90.86	84.61	87.58	71.25	68.39	75.82	73.07	68.06	78.73	51.01	48.87
	KFIoU	✓	89.09	83.71	55.21	82.81	80.18	84.11	88.60	90.86	85.01	87.69	68.89	69.60	73.40	79.67	68.19	79.13	44.02	45.35
	KFIoU∗	✓	89.18	83.16	56.87	80.14	80.26	84.18	88.83	90.83	84.43	87.67	68.97	73.62	75.51	79.76	67.57	79.40	47.49	46.49

4) Detectors: The RetinaNet-O produces a large performance decrease when n is less than 4, with unstable results in the [4, 6]. It gets a steady performance increase of about 1% when the range is [6, 10], after which the overall performance begins to decrease. In the range of [6, 10], the overall detection performance of the RetinaNet-O is improved by 0.89% and the detection performance of SARO is improved by 2.48%.

For R^3Det, the optimal range of n is [8, 12], the overall detection performance is improved by 0.76%, and the detection performance of SARO is improved by 1.65%.

D. More Experiments

1) Results on DOTA-v1.0 Trainval/Test: To get high-precision metrics for PL and BD, we use the train/val set of DOTA-v1.0 for training and testing, respectively, in Section V-C. To further validate the effectiveness of the proposed method, we use the trainval/test set of DOTA-v1.0 for training and testing, respectively, in this section. The result is shown in Table IV, where ∗ denotes applying shape balance strategy during training, and MS denotes that it uses multiscale training and testing. The backbone of RetinaNet-O and R^3Det is ResNet50. We set the parameter n corresponding to GWD and KLD losses to 8, while $n = 6$ for KFIoU. This configuration is also used in the related experiments on DIOR-R dataset (see Table V).

Applying the shape balance strategy improves the high-precision performance of the detector. For RetinaNet-O, the AP_{50} only was improved by 0.06%, but AP_{75} was improved by an average of 2.19%, and mAP was improved by an average of 1.04%. In the single-scale mode, the AP_{75} metric of the detector was improved by 2.36% and mAP by 1.11%. In the multiscale mode, the AP_{75} was improved by 2.02% and mAP by 0.97%. For R^3Det, the AP_{50} was improved by 0.13%, but AP_{75} was improved by an average of 1.97%, and mAP was improved by an average of 0.99%. In the single-scale mode, the AP_{75} metric of the detector was improved by 1.65% and mAP by 1.10%. In the multiscale mode, the AP_{75} was improved by 2.29% and mAP by 0.87%. We visualized some of these results, as shown in Fig. 6. The left side of the green vertical line results from the R^3Det with KFIoU, and the right side is the results after applying the shape balance strategy. SAROs with a well-defined orientation such as airplanes and BDs obtain remarkable improvements in precision.

We take R^3Det with GWD loss as a baseline and use the Eigen-CAM [20] tool to visualize and compare the feature maps applying the shape balance strategy (the second-last layer of the classification branch). The result is shown in Fig. 7. By Eigen-CAM [20] comparison, shape balance makes the model more focused on the area where the SARO is, which suppresses the impact of background noise in a certain degree, and the original GDL also focuses on the object but the highlighted red area that is most focused on does not cover the object completely.

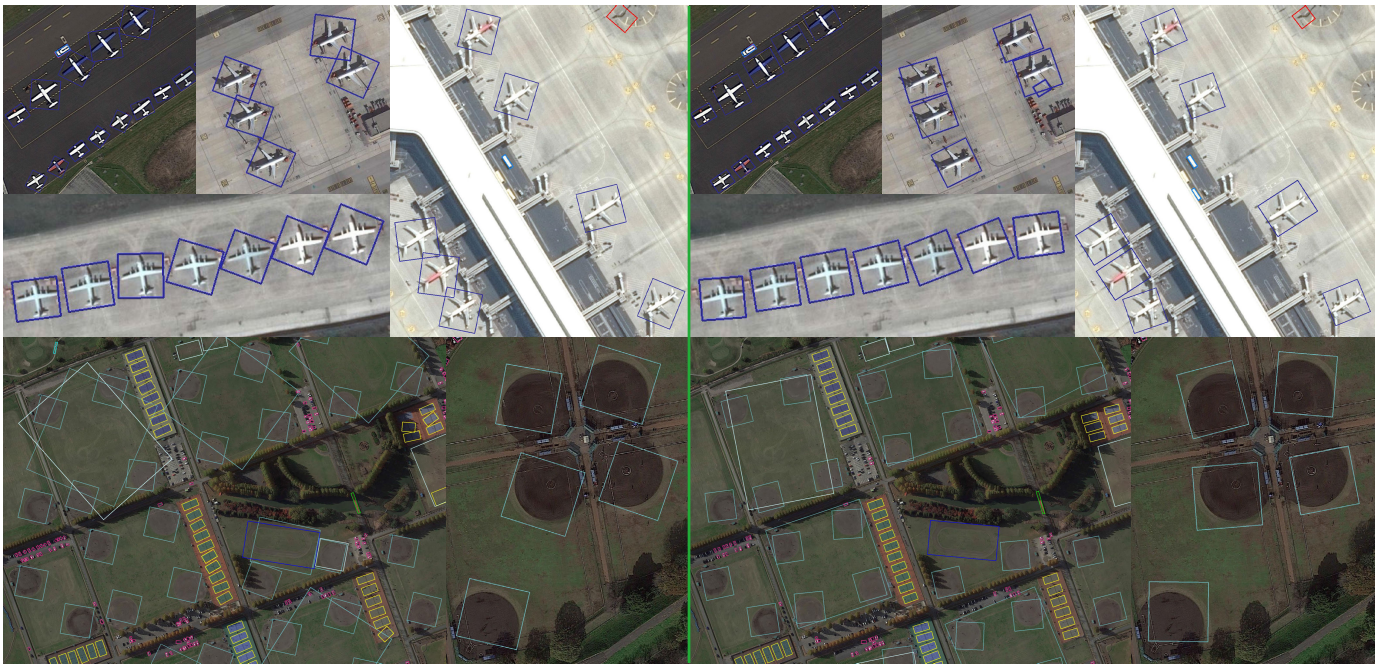
From the above results, the key to the overall detection performance improvement of the detector is the significant increase in the AP_{75} metric, while the low-precision metrics such as AP_{50} almost unchanged. The shape balance strategy aims to improve the high-precision performance of the detector by rebalancing the gradient gap between objects with different aspect ratios. This is consistent with our vision.

2) Results on DIOR-R: Considering the dataset bias, we also conducted relevant comparison experiments on DIOR-R data. The result is shown in Table V, where the meaning of the symbol ∗ is the same as in Table IV. Because the DIOR-R dataset provides the label for the test set, we can evaluate high-precision metrics for SARO.

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TABLE V
COMPARISON OF SHAPE BALANCE ON DIOR-R TRAINVAL/TEST SET (∗ MEANS USING SHAPE BALANCE)

Detectors	Loss	Airplane			Baseball Field			Total		
		AP_{50}	AP_{75}	mAP	AP_{50}	AP_{75}	mAP	AP_{50}	AP_{75}	mAP
RetinaNet-O[13]	KLD	61.40	20.21	29.87	73.37	43.77	46.21	55.55	33.52	33.65
	KLD∗	61.89 ↑0.48	23.32 ↑3.11	30.66 ↑0.79	73.79 ↑0.42	45.34 ↑1.57	48.30 ↑2.09	54.98 ↓0.57	34.28 ↑0.75	34.11 ↑0.46
	GWD	60.84	19.92	29.03	72.85	44.70	46.51	55.20	33.91	34.07
	GWD∗	59.98 ↓0.86	20.34 ↑0.42	30.12 ↑1.09	71.70 ↓1.15	45.35 ↑0.64	47.90 ↑1.39	55.13 ↓0.06	34.40 ↑0.49	34.50 ↑0.44
	KFIoU	62.01	19.24	29.78	69.94	43.01	45.01	54.46	29.55	31.10
	KFIoU∗	60.26 ↓1.75	20.75 ↑1.52	30.55 ↑0.77	71.68 ↑1.74	44.84 ↑1.82	46.23 ↑1.23	55.16 ↑0.70	31.01 ↑1.46	31.95 ↑0.86
	SGKLD	59.55	26.94	32.93	72.65	45.77	46.98	54.21	34.16	34.06
R^3Det[31]	KLD	52.77	28.82	30.50	70.44	47.96	46.67	58.43	36.33	35.63
	KLD∗	56.45 ↑3.68	29.77 ↑0.96	33.16 ↑2.66	71.03 ↑0.59	49.46 ↑1.50	48.79 ↑2.13	58.21 ↓0.21	37.50 ↑1.18	36.40 ↑0.77
	GWD	58.88	28.34	32.60	71.98	48.02	48.29	59.00	36.55	36.46
	GWD∗	59.95 ↑1.07	30.34 ↑1.99	35.15 ↑2.55	71.58 ↓0.40	50.61 ↑2.59	50.34 ↑2.05	58.72 ↓0.28	37.44 ↑0.89	36.67 ↑0.21
	KFIoU	52.30	23.26	29.33	68.67	47.60	44.78	56.44	30.87	32.18
	KFIoU∗	56.25 ↑3.95	28.89 ↑5.63	32.54 ↑3.21	69.33 ↑0.66	48.52 ↑0.93	46.66 ↑1.87	57.48 ↑1.04	32.91 ↑2.04	33.29 ↑1.11
	SGKLD	63.36	31.49	36.34	72.42	52.02	49.75	57.43	37.60	36.05
S^2ANet[7]	KLD	62.24	27.80	33.86	73.30	47.98	48.96	61.09	36.67	36.24
	KLD*	62.39 ↑0.14	29.42 ↑1.63	35.03 ↑1.17	73.17 ↓0.13	50.61 ↑2.62	49.12 ↑0.16	59.57 ↓1.53	37.27 ↑0.60	36.24 ↓0.01
	GWD	67.03	26.83	33.78	73.51	47.28	48.87	61.63	37.10	36.88
	GWD*	66.22 ↓0.81	28.44 ↑1.61	34.67 ↑0.89	73.86 ↑0.35	49.92 ↑2.63	49.66 ↑0.79	61.43 ↓0.20	37.55 ↑0.45	37.10 ↑0.22
	KFIoU	69.55	25.53	33.54	75.41	46.81	46.79	62.28	33.41	34.39
	KFIoU*	69.79 ↑0.24	27.85 ↑2.32	35.90 ↑2.36	74.55 ↓0.86	47.60 ↑0.78	47.94 ↑1.15	62.22 ↓0.07	34.30 ↑0.89	35.06 ↑0.66
	SGKLD	70.77	30.23	39.59	76.94	56.31	53.53	61.87	36.57	36.74
ROI Trans.[6]	KLD	63.14	41.43	40.91	71.97	58.96	53.44	63.53	43.37	41.03
	KLD*	63.21 ↑0.07	42.64 ↑1.21	42.23 ↑1.32	71.99 ↑0.02	59.61 ↑0.66	55.23 ↑1.78	63.44 ↓0.08	44.21 ↑0.83	41.74 ↑0.70
	GWD	63.08	40.87	39.82	71.98	58.22	52.29	63.76	43.08	40.94
	GWD*	63.24 ↑0.16	41.26 ↑0.39	41.13 ↑1.31	72.03 ↑0.05	58.75 ↑0.53	53.78 ↑1.49	62.86 ↓0.91	43.43 ↑0.36	41.19 ↑0.26
	KFIoU	63.20	39.90	39.25	71.98	57.27	52.18	63.17	40.70	38.88
	KFIoU*	63.15 ↓0.04	40.86 ↑0.96	40.23 ↑0.98	71.93 ↓0.06	58.79 ↑1.53	52.61 ↑0.43	63.21 ↑0.04	40.91 ↑0.21	39.22 ↑0.33
	SGKLD	63.06	43.18	41.62	71.85	59.96	55.89	62.38	42.52	40.44
RTMDet[19]	KLD	52.26	13.82	22.23	71.04	44.85	43.31	51.12	21.28	24.69
	KLD*	53.37 ↑1.11	14.88 ↑1.06	25.51 ↑3.29	72.57 ↑1.53	47.79 ↑2.94	46.03 ↑2.73	53.46 ↑2.34	24.56 ↑3.28	27.30 ↑2.61
	GWD	59.20	10.96	22.74	70.37	42.76	44.09	51.02	21.44	25.03
	GWD*	61.13 ↑1.93	13.42 ↑2.46	27.31 ↑4.58	69.74 ↓0.63	49.28 ↑6.53	46.38 ↑2.28	51.09 ↑0.06	23.56 ↑2.12	26.19 ↑1.16
	KFIoU	59.34	12.07	22.13	71.36	44.57	44.62	51.74	21.68	24.83
	KFIoU*	60.17 ↑0.83	14.34 ↑2.27	26.67 ↑4.54	70.83 ↓0.53	49.57 ↑4.99	47.05 ↑2.43	51.38 ↓0.36	23.73 ↑2.05	26.49 ↑1.66
	SGKLD	53.64	24.59	28.24	67.93	48.87	46.48	52.59	31.97	31.55



R^3Det with KFIoU R^3Det with KFIoU+Shape balance

Fig. 6. Comparison of visualization results of KFIoU and shape balance on DOTA-v1.0. The confidence threshold is set to 0.3 when visualizing these results.

Taking an overall view, SARO's AP_{75} metrics was improved by an average of 1.99% and mAP by 1.85%. The overall AP_{75} metrics was improved by 1.17% and mAP by 0.82%. Analyzing the results of AP_{75} and mAP, it can be obtained

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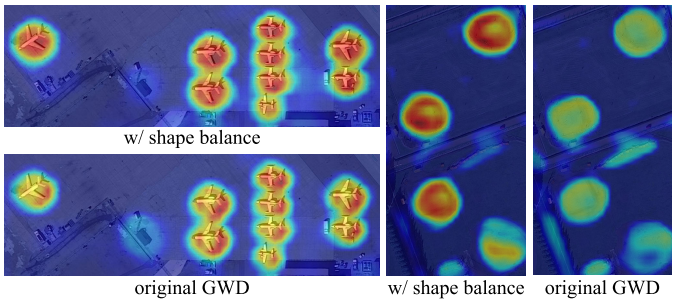


Fig. 7. Comparison of feature map Eigen-CAM [20] result between with (w/) and without (w/o) shape balance in GWD. The baseline detector is R^3Det, and the feature map is the second-last layer of the classification branch in the detection head.

that the shape balance strategy significantly improves the detection performance of SARO and it is more effective than the other categories. The shape balance performs unstable on low-precision metrics such as AP_{50}, while the high-precision metrics achieve a steady improvement, which is consistent with the situations in Tables I–IV. The shape balance is weaker in terms of SARO's detection performance improvement compared to the SGKLD but outperforms the SGKLD in terms of overall detection performance and low-precision metrics, such as AP_{50}. SGKLD shows a significant performance advantage over other GDLs for SARO, and the reasons SGKLD still works for SARO will be analyzed in Section V-D3.

3) Rethinking Shape Balance and SGKLD: The aspect ratio of many SAROs is not equal to 1. Such objects can still keep their angular information encoded by a Gaussian distribution, so the low SARO detection performance cannot be perfectly explained by the angular missing problem alone. As seen from Table V, the SGKLD loss based on the super-Gaussian distribution improves the detection performance of SARO. If only addressing the angle missing problem, SGKLD would not have such a significant performance improvement. Therefore, we performed the same gradient analysis (similar to Fig. 2) for SGKLD, and the results are shown in Fig. 8. As seen from Fig. 8, SGKLD reduces the gradient ratio by one order of magnitude compared with other GDL, which narrows the gradient gap between objects with different aspect ratios. It leads to the SGKLD still working for SAROs. While SGKLD reduces the gradient gap between SAROs and LAROs, the rise in the base rank brings an overall increase in the gradient as well, causing the gradient of LAROs to be too large and affecting the speed of convergence of the deep learning model.

Shape balance and SGKLD have similar goals, but they take two very different ways. SGKLD essentially changes the properties of the distribution, making the super-Gaussian representation of square-like objects with anisotropy, which solves the angular missing problem. On the other hand, shape balance achieves its goal by reinforcing the gradient of SAROs, essentially not eliminating the angle missing problem, but only mitigating the shape imbalance. However, the super-Gaussian distribution has many limitations, and it is difficult to extend it to Wasserstein distances and Bhattacharyya distances, etc., which makes it difficult to apply. In comparison, the shape balance strategy has wider applicability.

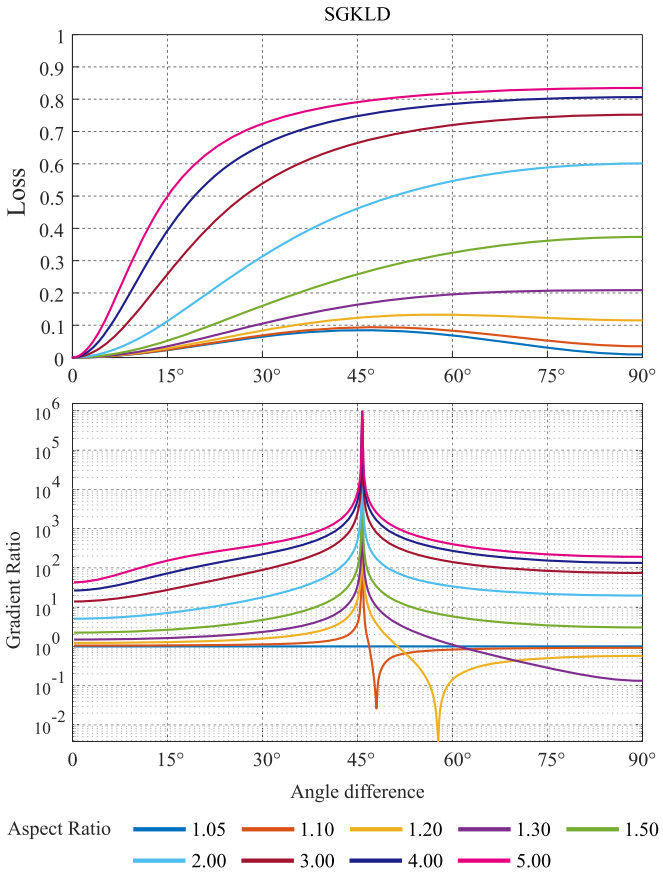


Fig. 8. Comparison of different aspect ratio objects on SGKLD and their gradients. Both loss and gradient are calculated by adjusting the angular difference between two objects with the same scale and center point. All gradients are divided by the gradient when the aspect ratio is 1.05, so it has a constant gradient ratio of 1. The SGKLD narrows the large gradient gap that exists in GDL.

Shape balance adjusts the gradients of objects with different aspect ratios to reduce the large gap that exists between them in a way that improves the high-precision metrics of the SAROs, but also has some impact on performing the LAROs. According to (4), the aspect ratio of the object is inversely proportional to the balance coefficient, so a larger aspect ratio means that the upper limit of the balance coefficient is smaller, and the effect on the gradient is smaller. According to (4), the shape balance does not significantly change the gradient of LAROs and has less effect on it. From the results in Table V, the average AP_{75} improvement for non-SAROs is 1.08%, while the average mAP improvement is 0.75%. In contrast, the average performance improvement for SAROs is 1.99% and 1.85%, which are 1.8 and 2.5 times higher than that for non-SAROs. From another perspective, the IoU in the shape balance strategy not only controls the period in which the balancing strategy takes effect but also allows the model to focus on the prediction points that work better under the same label, creating a competitive advantage that produces an effect similar to focal loss. Consequently, non-SAROs also demonstrated a small enhancement in detection performance.

SGKLD is a localization loss based on a super-Gaussian distribution. For super-Gaussian distribution, there is difficulty in computing analytical solutions for probability-distributed distances, except KL divergence. It makes the super-Gaussian

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distribution limited in its current applications. Second, as the rank increases, it also brings about a rapid increase in gradient. It does not favor the learning of LARO's boundary information. The shape balance strategy relies on hand-designed balance functions, and the hyperparameters of the functions need to be carefully discussed, making shape balance limited. Future study work can be in the following two directions, exploring new probability distributions and designing adaptive gradient balancing mode. SGKLD constructs a non-Gaussian distribution localization loss. Therefore, different probability distributions can also be explored for the properties of oriented objects to design a novel localization loss. Self-dynamically balance the localization loss and gradient by introducing the pattern of focal loss is also an improvement idea.

VI. Conclusion

GWD first employed a 2-D Gaussian distribution to represent the OBBs, and other classical GDLs such as ProbIoU, KLD, and KFIoU were successively proposed. They show great power for general oriented objects, but difficulty in predicting the angles of SAROs and square-like objects. Square-like objects suffer from the problem of angle loss because of isotropy Gaussian distributions. SAROs are affected by the phenomenon of shape imbalance, which results in large gaps in angular gradients between objects with different aspect ratios. We propose a simple gradient rebalancing strategy to narrow the gap between them. Extensive experiments on multiple detectors and GDL confirm the effectiveness of the proposed method. Overall, shape balance can stably improve the detection performance of Gaussian representations for SARO.

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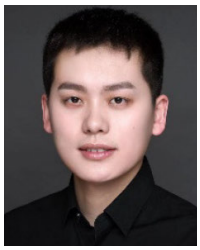
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